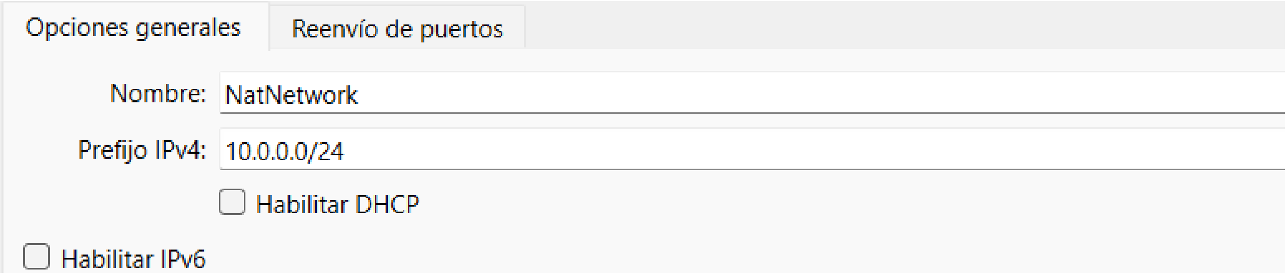
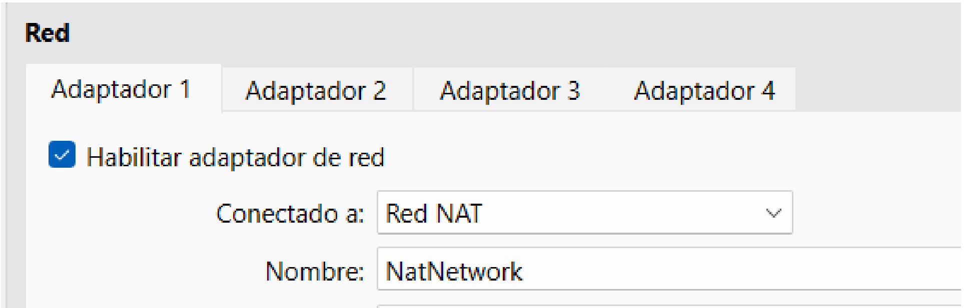
Creamos una red nat

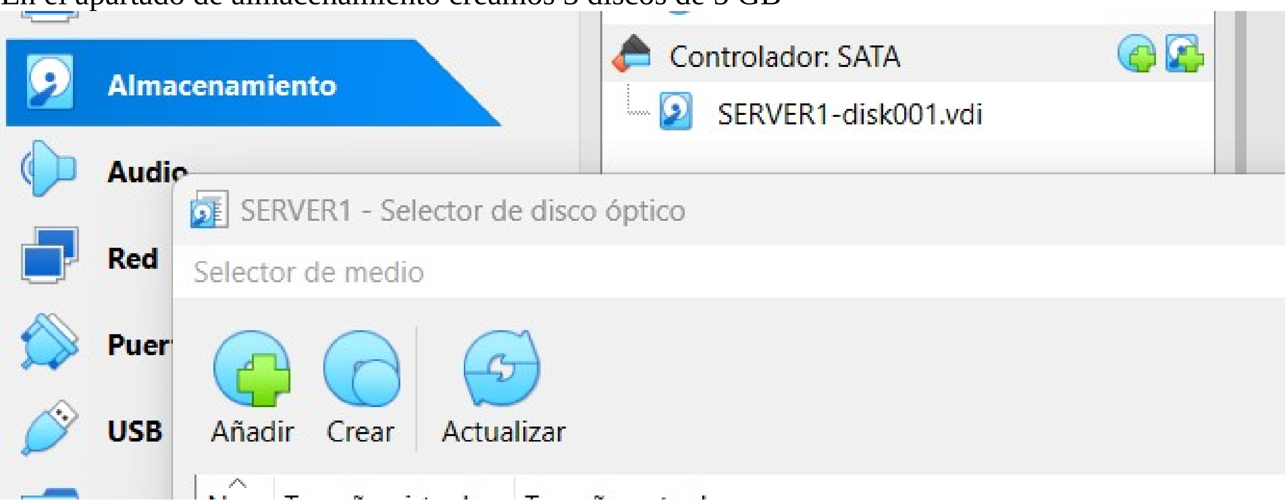


Y se la aplicamos a todas las MV

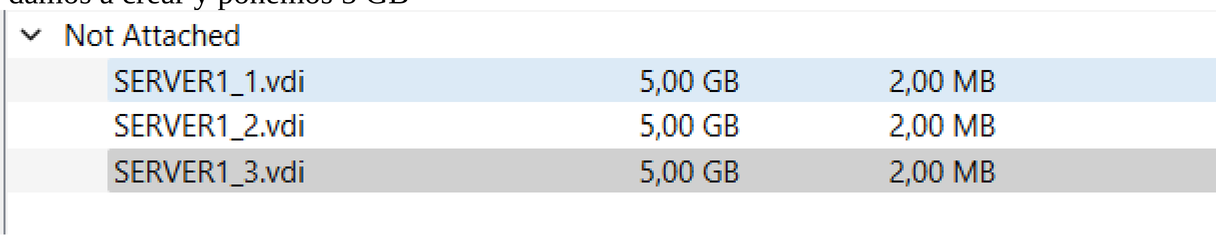


a.

En el apartado de almacenamiento creamos 3 discos de 5 GB



Le damos a crear y ponemos 5 GB



Y para añadirlos le damos doble click

Comprobamos que se han añadido correctamente con un lsblk

```
alumno@server1:~$ lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                       7:0    0 38,8M  1 loop /snap/snapd/21759
loop1                       7:1    0   87M  1 loop /snap/lxd/29351
loop2                       7:2    0 63,9M  1 loop /snap/core20/2318
sda                         8:0    0   25G  0 disk
├─sda1                      8:1    0    1M  0 part
├─sda2                      8:2    0    2G  0 part /boot
└─sda3                      8:3    0   23G  0 part
  └─ubuntu--vg-ubuntu--lv 253:0    0 11,5G  0 lvm  /
sdb                         8:16   0    5G  0 disk
sdc                         8:32   0    5G  0 disk
sdd                         8:48   0    5G  0 disk
sr0                        11:0    1 1024M  0 rom
```

Ahora creamos particiones gpt de 2,5 GB
con este comando

```
alumno@server1:~$ sudo fdisk /dev/sdd

Welcome to fdisk (util-linux 2.37.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): _
```

Escribimos g para una gpt

```
Command (m for help): g
Created a new GPT disklabel (GUID: F4675E09-21B4-6B4C-91E4-DD7BDDADE158).

Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-10485726, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485726, default 10485726):
```

En tamaño ponemos +2,5G, y despues w para guardar

Así en todos y vemos que se ha guardado

```
sdb                         8:16   0    5G  0 disk
└─sdb1                      8:17   0  2,5G  0 part
sdc                         8:32   0    5G  0 disk
└─sdc1                      8:33   0  2,5G  0 part
sdd                         8:48   0    5G  0 disk
└─sdd1                      8:49   0  2,5G  0 part
sr0                        11:0    1 1024M  0 rom
alumno@server1:~$
```

c.
Para otorgarle el sistema de archivos ext4 escribimos esto

```
alumno@server1:~$ sudo mkfs.ext4 /dev/sdb
mke2fs 1.46.5 (30-Dec-2021)
Found a gpt partition table in /dev/sdb
Proceed anyway? (y,N) y
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 4f91c7c5-e72d-45d7-953b-89be520bb3b8
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

le damos a y
Y asi con todos

```
alumno@server1:~$ sudo mkfs.ext4 /dev/sdc
mke2fs 1.46.5 (30-Dec-2021)
Found a gpt partition table in /dev/sdc
Proceed anyway? (y,N) y
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 0beb3654-f8a6-4962-985e-9c6b1f2144ab
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

alumno@server1:~$ sudo mkfs.ext4 /dev/sdd
mke2fs 1.46.5 (30-Dec-2021)
Found a gpt partition table in /dev/sdd
Proceed anyway? (y,N) y
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: e12a4ecd-7ae7-46f0-b0fc-36bfcbf437ef
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

d.

Creamos la carpeta

```
alumno@server1:~$ sudo mkdir -p carpeta1
```

Montamos en la carpeta

```
alumno@server1:~$ sudo mount /dev/sdb ~/carpeta1
```

Nos movemos

```
alumno@server1:~$ cd carpeta1
alumno@server1:~/carpeta1$ _
```

Y creamos y editamos un archivo

```
  GNU nano 6.2                                          1.txt *
hola_
```

Vemos que se ha creado

```
alumno@server1:~/carpeta1$ ls
1.txt  lost+found
```

Y desmontamos y vemos que no esta el archivo al desmonarlo

```
alumno@server1:~$ sudo umount /dev/sdb ~/carpeta1
umount: /home/alumno/carpeta1: not mounted.
alumno@server1:~$ ls
carpeta1  carpeta2
alumno@server1:~$ _
```

```
alumno@server1:~$ cd carpeta1
alumno@server1:~/carpeta1$ ls
```

Lo mismo con los otros dos discos

```
alumno@server1:~$ sudo mount /dev/sdc ~/carpeta1
alumno@server1:~$ cd carpeta1
alumno@server1:~/carpeta1$ sudo nano 2.txt
```

```
  GNU nano 6.2                                          2.txt *
hola2_
```

```
alumno@server1:~/carpeta1$ ls
2.txt   lost+found
alumno@server1:~/carpeta1$ sudo umount /dev/sdc ~/carpeta1
umount: /home/alumno/carpeta1: target is busy.
umount: /home/alumno/carpeta1: target is busy.
alumno@server1:~/carpeta1$ cd ..
alumno@server1:~$ sudo umount /dev/sdc ~/carpeta1
umount: /home/alumno/carpeta1: not mounted.
```

```
alumno@server1:~$ ls
carpeta1  carpeta2
alumno@server1:~$
```

```
alumno@server1:~$ sudo mount /dev/sdd ~/carpeta1
alumno@server1:~$ cd carpeta1
alumno@server1:~/carpeta1$ sudo nano 3.txt_
```

```
  GNU nano 6.2                                               3.txt *
hola3_
```

```
alumno@server1:~/carpeta1$ ls
3.txt   lost+found
alumno@server1:~/carpeta1$
```

```
alumno@server1:~/carpeta1$ cd ..
alumno@server1:~$ sudo umount /dev/sdd ~/carpeta1
umount: /home/alumno/carpeta1: not mounted.
alumno@server1:~$ ls
carpeta1  carpeta2
alumno@server1:~$ cd carpeta1
alumno@server1:~/carpeta1$ ls
alumno@server1:~/carpeta1$
```

2.

a. b.

Para crear el raid escribimos esto

```
alumno@server1:~$ sudo mdadm --create /dev/md0 --level=5 --raid-devices=3 /dev/sdb /dev/sdc /dev/sdd
mdadm: /dev/sdb appears to contain an ext2fs file system
       size=5242880K  mtime=Thu Mar  5 09:41:12 2026
mdadm: /dev/sdc appears to contain an ext2fs file system
       size=5242880K  mtime=Thu Mar  5 09:44:59 2026
mdadm: /dev/sdd appears to contain an ext2fs file system
       size=5242880K  mtime=Thu Mar  5 09:46:29 2026
Continue creating array? y
mdadm: Defaulting to version 1.2 metadata
mdadm: array /dev/md0 started.
```

c.

Le damos el sistema de archivos

```
alumno@server1:~$ sudo mkfs.ext4 /dev/md0
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2618880 4k blocks and 655360 inodes
Filesystem UUID: 2697d0a7-edf9-4c06-999a-b0149eb45cdc
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

d. e.

Creamos el directorio

```
alumno@server1:~$ sudo mkdir -p RAID5
alumno@server1:~$ _
```

Montamos el raid

```
alumno@server1:~$ sudo mount /dev/md0 ~/RAID5
```

Y creamos el archivos

```
alumno@server1:~$ cd RAID5
alumno@server1:~/RAID5$ sudo nano raid.txt
```

```
  GNU nano 6.2                                  raid.txt *
raid hola_
```

f.

Hacemos un lsblk -f para ver el UUID del raid y copiarlo

```
alumno@server1:~$ lsblk -f
NAME   FSTYPE FSVER LABEL     UUID                                   FSAVAIL FSUSE% MOUNTPOINTS
loop0  squash 4.0                                                          0   100% /snap/snapd/21759
loop1  squash 4.0                                                          0   100% /snap/lxd/29351
loop2  squash 4.0                                                          0   100% /snap/core20/2318
sda
├─sda1
│
├─sda2
│      ext4   1.0             3f13938b-a8de-4496-8649-d26eb1cd7590      1,7G     7% /boot
└─sda3
       LVM2_m LVM2            woLfcE-Lhv6-8gHS-Ej71-Rca8-sish-YJnRdf
  └─ubuntu--vg-ubuntu--lv
       ext4   1.0             09404c95-fdff-4782-9677-372234d9c532      5,4G    47% /
sdb    linux_ 1.2   server1:0
│                             d6c579c6-a2f2-5cc0-d938-b5b2584b3cfb
└─md0  ext4   1.0             2697d0a7-edf9-4c06-999a-b0149eb45cdc      9,2G     0% /home/alumno/RAID5
sdc    linux_ 1.2   server1:0
│                             d6c579c6-a2f2-5cc0-d938-b5b2584b3cfb
└─md0  ext4   1.0             2697d0a7-edf9-4c06-999a-b0149eb45cdc      9,2G     0% /home/alumno/RAID5
sdd    linux_ 1.2   server1:0
│                             d6c579c6-a2f2-5cc0-d938-b5b2584b3cfb
└─md0  ext4   1.0             2697d0a7-edf9-4c06-999a-b0149eb45cdc      9,2G     0% /home/alumno/RAID5
sr0
alumno@server1:~$
```

```
└─md0 ext4   1.0             2697d0a7-edf9-4c06-999a-b0149eb45cdc
```

Y lo escribimos junto a la ruta en el fstab

```
  GNU nano 6.2                                    /etc/fstab *
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
# / was on /dev/ubuntu-vg/ubuntu-lv during curtin installation
/dev/disk/by-id/dm-uuid-LVM-uinaW3jKQerN71lpdKwOKMgWLKkT25LfFxm3bNZrQivOf9EKOdXbtwRbWuf3
# /boot was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/3f13938b-a8de-4496-8649-d26eb1cd7590 /boot ext4 defaults 0 1
/swap.img       none    swap    sw      0       0
uuid=2697d0a7-edf9-4c06-999a-b0149eb45cdc /home/alumno/RAID5 ext4 defaults 0 2
```

3.
a. b.

Creamos el directorio

```
alumno@server1:~/RAID5$ sudo mkdir -p compartida/clase/alumnos
```

Y los dos documentos de prueba comprobando que se crean correctamente

```
alumno@server1:~/RAID5$ cd /compartida/clase/alumnos
-bash: cd: /compartida/clase/alumnos: No such file or directory
alumno@server1:~/RAID5$ cd compartida/clase/alumnos
alumno@server1:~/RAID5/compartida/clase/alumnos$ sudo touchv doc1.txt
sudo: touchv: command not found
alumno@server1:~/RAID5/compartida/clase/alumnos$ sudo touch doc1.txt
alumno@server1:~/RAID5/compartida/clase/alumnos$ sudo touch doc2.txt
alumno@server1:~/RAID5/compartida/clase/alumnos$ ls
doc1.txt  doc2.txt
```

` Ahora editamos la configuracion de red

```
 GNU nano 6.2                      /etc/netplan/50-cloud-init.yaml
network:
   ethernets:
       enp0s3:
           dhcp4: false
           addresses: [10.0.0.10/24]
           routes:
             - to: default
               via: 10.0.0.1
           nameservers:
              addresses: [8.8.8.8, 8.8.4.4]
   version: 2
```

```
 GNU nano 6.2                      /etc/netplan/50-cloud-init.yaml
network:
   ethernets:
       enp0s3:
           dhcp4: true
           addresses: [10.0.0.20/24]
           routes:
             - to: default
               via: 10.0.0.1
           nameservers:
              addresses: [8.8.8.8, 8.8.4.4]
   version: 2
```

```
alumno@server2:~$ ping 10.0.0.10
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.
64 bytes from 10.0.0.10: icmp_seq=1 ttl=64 time=2.39 ms
64 bytes from 10.0.0.10: icmp_seq=2 ttl=64 time=3.07 ms
^C
--- 10.0.0.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.386/2.726/3.067/0.340 ms
```

```
alumno@server1:~/RAID5/compartida/clase/alumnos$ ping 10.0.0.20
PING 10.0.0.20 (10.0.0.20) 56(84) bytes of data.
64 bytes from 10.0.0.20: icmp_seq=1 ttl=64 time=1.71 ms
^C
--- 10.0.0.20 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.711/1.711/1.711/0.000 ms
alumno@server1:~/RAID5/compartida/clase/alumnos$
```

c.

Ahora desde el server1 entramos a /etc/exports y editamos esto

```
  GNU nano 6.2                                    /etc/exports *
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/home/alumno/RAID5/compartida/clase/alumnos 10.0.0.20(rw,sync,no_subtree_check)
```

Y exportamos con este comando

```
alumno@server1:~$ sudo exportfs -av
exporting 10.0.0.20:/home/alumno/RAID5/compartida/clase/alumnos
```

Le damos permisos completos a la carpeta alumnos

```
alumno@server1:~$ sudo chmod 777 /home/alumno/RAID5/compartida/clase/alumnos
alumno@server1:~$ sudo chown alumno:alumno /home/alumno/RAID5/compartida/clase/alumnos
```

d.

Ahora en el server2 creamos un directorio y montamos por nfs a ese directorio

```
alumno@server2:~$ sudo mkdir comp
alumno@server2:~$ sudo mount -t nfs 10.0.0.10:/home/alumno/RAID5/compartida/clase/alumnos ~/comp
```

Cambiamos al directorio y con un ls vemos que estan los dos documentos de prueba

```
alumno@server2:~$ cd comp
alumno@server2:~/comp$ ls
doc1.txt  doc2.txt
```

4.

a.

Creamos los usuarios juan, maria y pepe

```
alumno@server1:~$ sudo adduser juan
Adding user `juan' ...
Adding new group `juan' (1001) ...
Adding new user `juan' (1001) with group `juan' ...
Creating home directory `/home/juan' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juan
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
```

```
alumno@server1:~$ sudo adduser maria
Adding user `maria' ...
Adding new group `maria' (1002) ...
Adding new user `maria' (1002) with group `maria' ...
The home directory `/home/maria' already exists.  Not copying from `/etc/skel'.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for maria
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
alumno@server1:~$
```

```
alumno@server1:~$ sudo adduser pepe
Adding user `pepe' ...
Adding new group `pepe' (1003) ...
Adding new user `pepe' (1003) with group `pepe' ...
Creating home directory `/home/pepe' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for pepe
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
```

b.

Creamos las carpetas para cada usuario

```
~# sudo mkdir /home/juan/carpeta1
~# sudo mkdir /home/maria/carpeta2
~# sudo mkdir /home/pepe/carpeta3
```

c.

Damos permisos

```
root@server1:~# sudo chmod 777 /home/pepe/carpeta3
root@server1:~# sudo chmod 777 /home/maria/carpeta2
root@server1:~# sudo chmod 777 /home/juan/carpeta1
```

```
root@server1:~# sudo chown juan:juan /home/juan/carpeta1
root@server1:~# sudo chown maria:maria /home/maria/carpeta2
root@server1:~# sudo chown pepe:pepe /home/pepe/carpeta3
root@server1:~#
```